

6.2.1 Chiller Maintenance. Services include to perform preventative work on the chillers, in the units themselves to repair any damage that might occur and flushing of the system when necessary. The contractor shall accomplish any work to maintain, flush and perform any preventative maintenance to maintain the system for optimal performance during the period of performance. There is regular maintenance that must be performed in order to maintain the equipment so that it operates optimally for the veterans that we serve and the staff that serve the Veteran population. This service is integral for the medical center to provide a comfortable environment for the different building and the areas that require a cooler temperature for medical procedures. The maintenance is beyond the scope of the A/C mechanics on station and therefore we are requesting that a contractor to perform the preventative maintenance and service.

6.2.1.1 The contractor shall perform maintenance services based upon the fault report noted by the COR or Duty Engineer. The contractor will receive a call from the CMC VAMC staff to request service and indicate what are the issues noted on the unit's display.

6.2.1.1.1 The contractor shall provide the contracting officer's representative (COR) an electronic copy of the checklist with all the information.

6.2.1.1.2 Manual records are not acceptable. Both the checklist and the service reports are to be provided electronically to the COR.

6.2.1.1.3 The diagnostic reports shall indicate all alarms failures, operating hours, and history. The Government requires all history since the last visit was performed. Full disclosure of all alarms, warnings, conditions within spec and out of spec shall be included. Chillers are protected by oil warning system and lubrication system which will shut off the chiller at the predetermined hour. Resetting of the chillers is included in this purchase order. The contractor shall advise the COR of all necessary corrections prior to commencing the repair. This repair may be handled in accordance with paragraph 6.1.4 of this document. Maintenance includes emergency responses 24 hours per day, seven days per week.

6.2.1.2 All listed chillers must be repaired and serviced by commercially certified technicians in accordance with manufacturers specifications to include technician certification requirements. These certifications are required as described in paragraph 6.3 below.

6.2.2 List of units located at 3900 Woodland Ave and 3701 Civic Center Boulevard, Philadelphia, PA 19104

Building	Quantity	Model	VSD	Installation Year
2	2	York 276-ton air cooled 480 volt	Yes	2011
2	1	York YCIVO177	Yes	2015
21	1	York 600-ton 480 volt	Yes	2011
30	2	York AB AB P2 300-ton 480 volt	Yes	2000

31	3	York 1200-ton 4160 volt	Yes	2011
31	1	York JHJB 1130-ton 480volt	Yes	2001

6.2.2.1 The Chiller building is located next to Building 2 of the CMC VAMC and is along the access road for easy access.

6.2.2.2 Building #2 has two, air-cooled, 267-ton capacity York chillers that serve the operating rooms. These are critical chillers to CMC VAMC. Both chillers supply air to the OR rooms and support areas on the 5th floor and the towers are located on the roof of Building 2. The frequency of all visits required shall be the same as the chillers, noting that these are air cooled units, York Model YCIV0267. Full services are required in accordance with the information in this package. Also building #2 has Lab service (AC19) York chiller with low voltage variable speed drive. Capacity is 177-ton, model # YCIV0177 with 2 compressors attached to this system. All warranty requirements shall apply for this chiller. All visits as noted, Eddy current testing yearly, oil renewal as noted, leak checking with refrigerant program as noted, all tube cleaning of the condenser tubes yearly for the chillers in building 2 at the end of the season. The warranty for A/C19 air cooled chiller will be void unless refrigerant, fan oil, variable speed drive parts are provided through York International. The parts and materials for all chillers are available through the York International parts location in Baltimore, Maryland. Maintenance performed contrary to York recommendations will void all warranties in place. The contractor shall be responsible for the terms of the existing agreement.

6.2.2.3 Research Building #21 has a York Centrifugal chiller with a low voltage variable frequency drive, the chiller is in the basement of Building 21 and the tower is located on the rooftop. Capacity is just under 600 tons. Plant consists of all pumps, towers, strainers, starter etc. This unit is a York Model YK and all warranty requirements shall apply. All visits as noted, Eddy current testing yearly, oil renewal as noted, leak checking with refrigerant program as noted, all tube cleaning of the condenser tubes yearly for the chiller in the building 21 at the end of the season.

6.2.2.4 CLC building #30 consists of two, York 300-ton capacity chiller model YKABABP2-CKF which shall receive all preventative maintenance tasks as noted chillers plant consists of all pumps, towers, strainers, starters piping, etc. The chiller units are in the basement of building 30 and the towers are located on the rooftop of the CLC. All visits as noted, Eddy current testing yearly, oil renewal as noted, leak checking with refrigerant program as noted, all tube cleaning of the condenser tubes yearly for the chillers in building 30 at the end of the season.

6.2.2.5 Building 31 consists of three York large tonnage centrifugal chillers 1200-ton capacity 4160 Volt remote mounted Variable Frequency Drives Toshiba with full services. These chillers are Model YK by York. The fourth unit is in Building #31 an earlier model design all at the same capacity of 1200-480 Volt tons for a 4800-ton approximate plant capacity. Plant consists of all pumps, towers, strainers, starters etc. The fourth unit has a unit mounted 4160 MVVSD and all necessary maintenance

shall be performed by factory-trained mechanics. The fourth chiller is a model YKJHJB1DAE. The contractor shall provide preventative maintenance tasks noted as well as the unit mounted MVVSD. All visits as noted, Eddy current testing yearly, oil renewal as noted, leak checking with refrigerant program as noted, all tube cleaning of the condenser tubes yearly for the chillers in building 30 at the end of the season. The York 4160 medium voltage variable speed drives (MVVSD) are manufactured by Toshiba. The MVVSD shall be serviced by technicians described in paragraph 6.3 of the SOW.

6.2.3 The contractor shall conduct an analysis of each centrifugal chiller annually comparing it to the manufacturer's specification for all energy efficiency data and report and note any corrective actions necessary. Report to be provided at no additional charge.

6.2.4 All chillers will receive the following inspections (Associated checklists are found in paragraph 6.9 below):

Quantity	Model	Comprehensive Service	Operational Service	Oil Testing/Changing	Tube Cleaning/coil cleaning
1	YCIV0177	1	6	1	1
2	YCIV0267	1	6	1	1
1	WC, centrifugal, 451-600T/Comp	1	6	1	1
3	WC, centrifugal, 451-1500T/Comp	1	6	1	1
1	YKJHJB1DAE	1	6	1	1
2	YKABABP2-CKF	1	6	1	1

6.2.5 Maintenance of each chiller includes the following as noted above:

6.2.5.1 Start-up for the cooling season;

6.2.5.2 Shut down for the heating season;

6.2.5.3 Operation and inspection during cooling season one per month;

6.2.5.4 Winter maintenance, to include cleaning of the tubes at the end of the season.

6.2.5.5 Eddy current testing for chiller condensers and evaporative testing, annually.

B.2 PRICE/COST SCHEDULE**ITEM INFORMATION**

ITEM NUMBER	DESCRIPTION OF SUPPLIES/SERVICES	QUANTITY	UNIT	UNIT PRICE	AMOUNT
0001	Wilmington VAMC Preventive Maintenance on Five (5) York / Carrier Chillers IAW SOW. Contract Period: Base POP Begin: POP End:	0.00		XXXXX	Do Not Quote
0001AA	Quarterly Preventive Maintenance on Five (5) York / Carrier Chillers IAW SOW. Wilmington VAMC. Contract Period: Base POP Begin: POP End:	4.00	JB		
0001AB	Annual Preventive Maintenance on Five (5) York / Carrier Chillers IAW SOW. Wilmington VAMC. Contract Period: Base POP Begin: POP End:	1.00	JB		
0002	Lebanon VAMC Preventive Maintenance on Two (2) Carrier Chillers IAW SOW. Contract Period: Base POP Begin: POP End:	0.00		XXXXX	Do Not Quote
0002AA	Annual Inspections on Two (2) Carrier Chillers IAW SOW. Lebanon VAMC. Contract Period: Base POP Begin: POP End:	1.00	JB		
0002AB	Operating Inspections on Two (2) Carrier Chillers IAW SOW. Lebanon VAMC. Contract Period: Base POP Begin: POP End:	2.00	JB		
0002AC	Condenser Cleaning on Two (2) Carrier Chillers IAW SOW. Lebanon VAMC.	2.00	JB		

Contract Period: Base POP Begin: POP End:				
0003	0.00	XXXXX	Do Not Quote	
CMC (Philadelphia) VAMC. Maintenance, Inspection, Repair and Testing on Ten (10) York Chillers IAW. Contract Period: Base POP Begin: POP End:				
0003AA	12.00	MO		
Monthly Maintenance, Inspection, Repair and Testing on Ten (10) York Chillers IAW. CMC VAMC.				
0003AB	0.00	EA	Government will provide dollar amount upon award	Do Not Quote
Emergency Repair \$2,500- \$15,000 IAW SOW. CMC VAMC. Contract Period: Base POP Begin: POP End:				
0003AC	0.00	EA	Government will provide dollar amount upon award	Do Not Quote
Parts or Repair over \$15,000 IAW SOW. CMC VAMC. Contract Period: Base POP Begin: POP End:				
0003AD	0.00	EA		
Additional Cleaning IAW SOW. CMC VAMC. Contract Period: Base POP Begin: POP End:				
GRAND TOTAL				

B.3 DELIVERY SCHEDULE

ITEM NUMBER	QUANTITY	DELIVERY DATE
0001	SHIP TO: Department of Veterans Affairs	0.00
	██████████	
	██████████████████	
	██████████████████████████████	
	USA	

0001AA	SHIP TO:	Department of Veterans Affairs [REDACTED] USA	4.00
0001AB	SHIP TO:	Department of Veterans Affairs [REDACTED] [REDACTED] [REDACTED] USA	1.00
0002	SHIP TO:	Department of Veterans Affairs [REDACTED] USA	0.00
0002AA	SHIP TO:	Department of Veterans Affairs [REDACTED] USA	1.00
0002AB	SHIP TO:	Department of Veterans Affairs [REDACTED] USA	2.00
0002AC	SHIP TO:	Department of Veterans Affairs [REDACTED] USA	2.00
0003	SHIP TO:	Department of Veterans Affairs [REDACTED] [REDACTED] [REDACTED] [REDACTED] USA	0.00
0003AA	SHIP TO:	Department of Veterans Affairs [REDACTED]	12.00

USA			
0003AB	SHIP TO:	Department of Veterans Affairs [REDACTED] [REDACTED] [REDACTED] [REDACTED] USA	0.00
0003AC	SHIP TO:	Department of Veterans Affairs [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]	0.00
0003AD	SHIP TO:	Department of Veterans Affairs [REDACTED] [REDACTED] [REDACTED] [REDACTED] USA	0.00